



experience molecular dynamics with a force-feedback joystick

Stefan Huber ¹ Pietro Sillano ² Timon Idema ²

¹ Visualisation & XR Team, SURF, Amsterdam

² Kavli Institute of Nanoscience Delft, Bionanoscience Department, TU Delft

Immersive Molecular Dynamics

In molecular dynamics simulations, forces and torques result in particle motion. The resulting dynamics are usually visualized as movies or a plots, something to watch rather than to play with.

In this interactive demo you can not only control the simulation, but also experience what happens in the simulation in an immersive manner, feeling the forces at work.

The MesoMem model

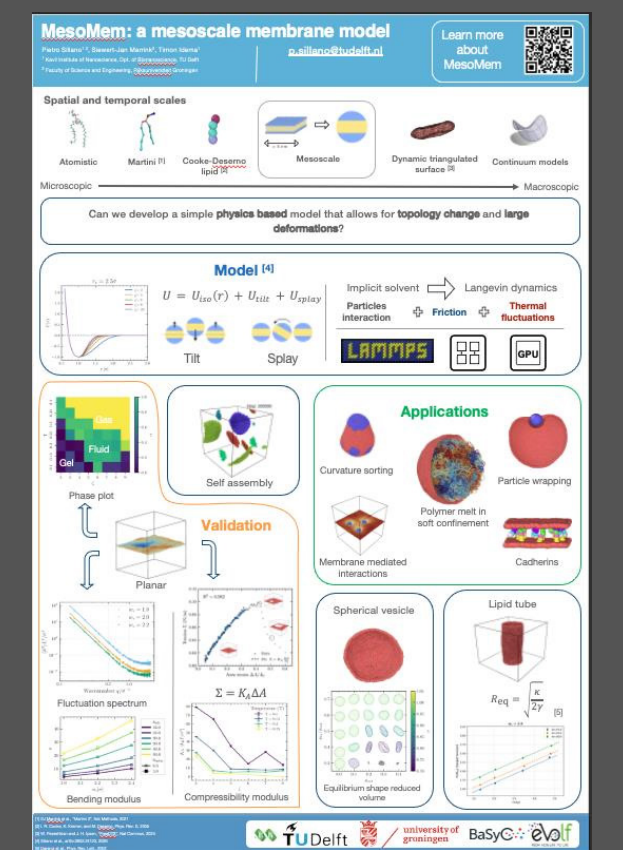
Mesomem is a solvent-free coarse-grained model of a biological lipid-bilayer membrane.

A patch of many lipids becomes one anisotropic bead, so the simulation can reach the length and time scales on which membranes self-assemble and remodel. That economy also makes it fast enough to run live.

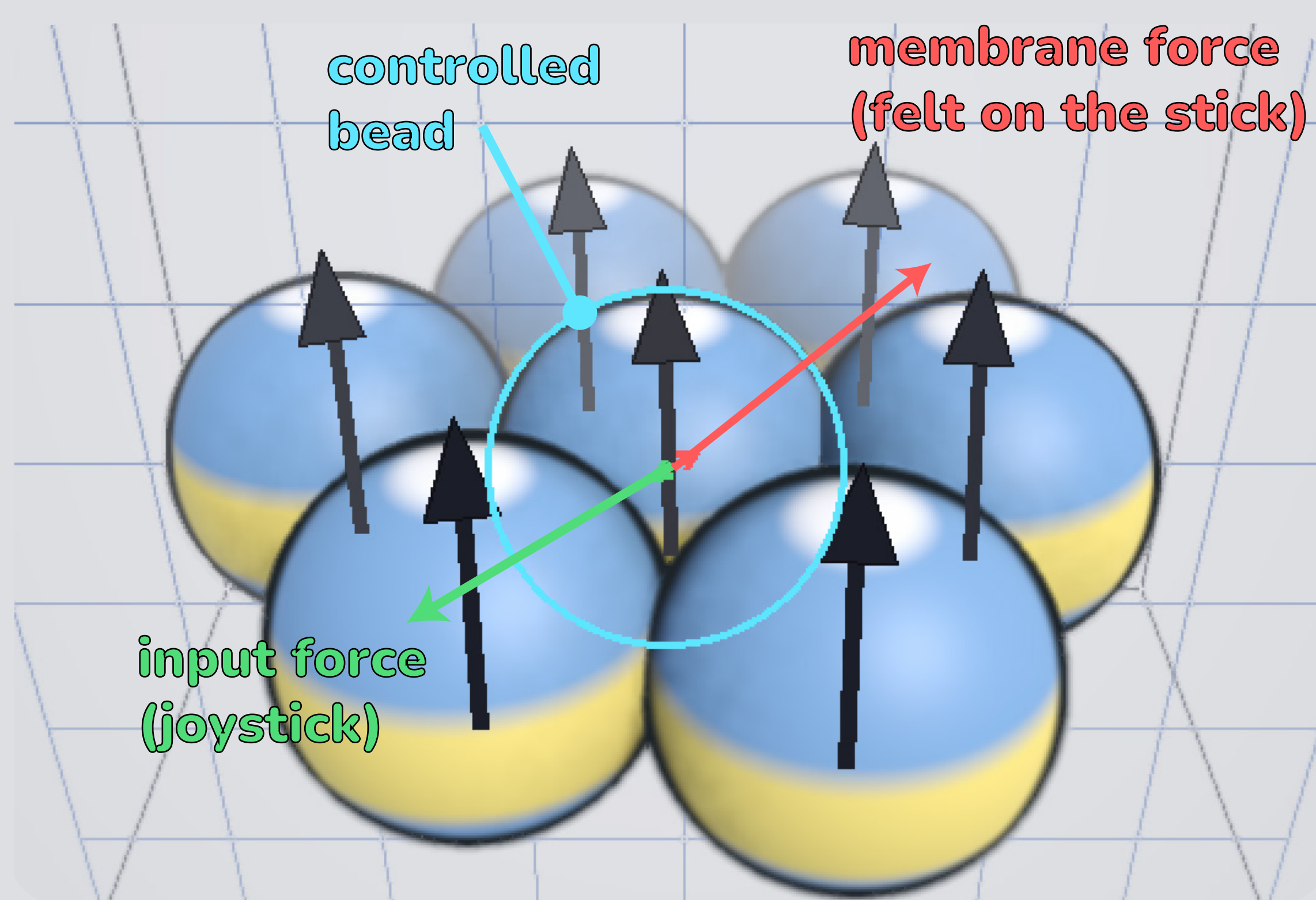
Three energy terms



see poster by Sillano et al.



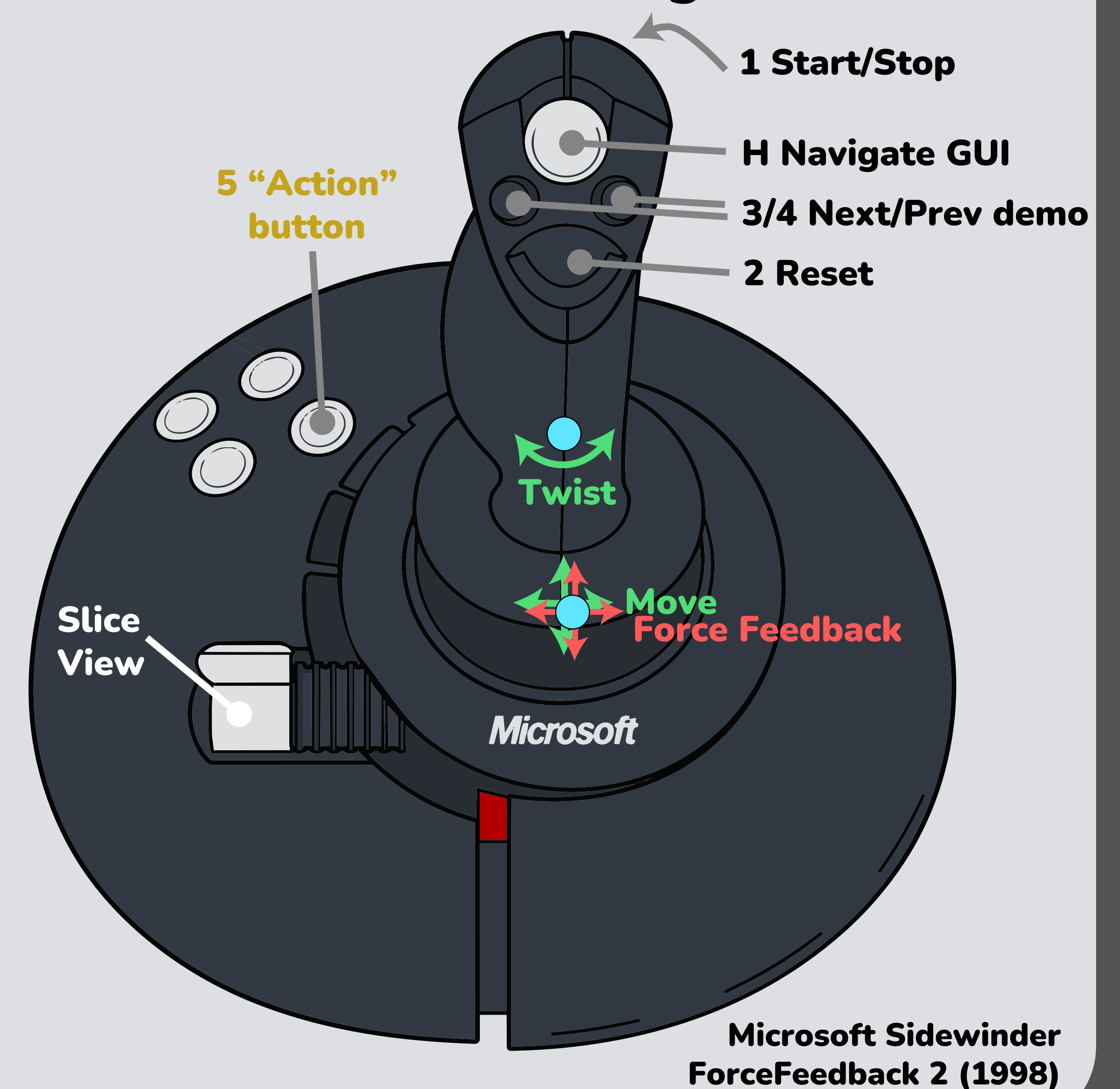
You control one bead.



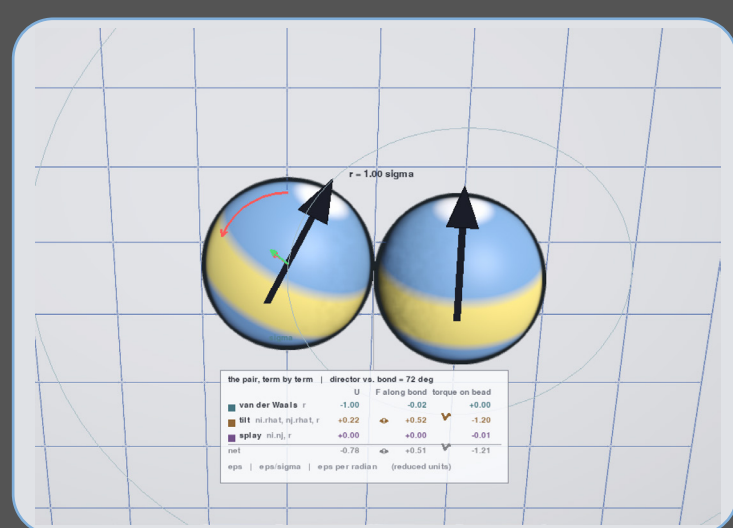
You can change the physics live



You feel the force acting on it.

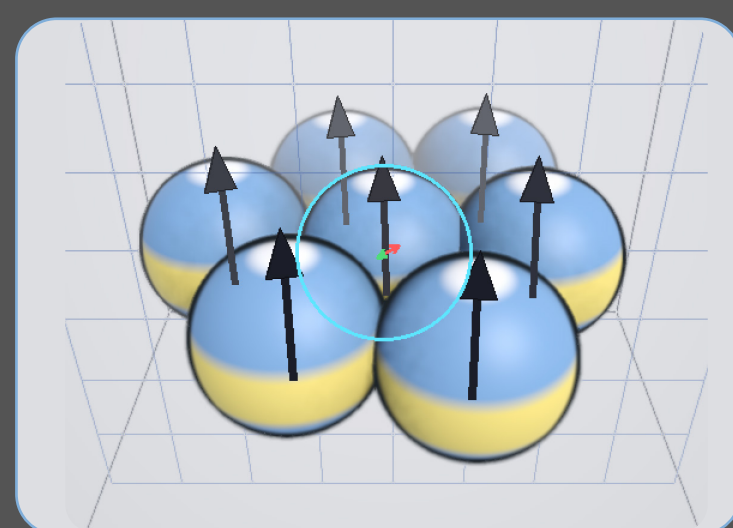


1



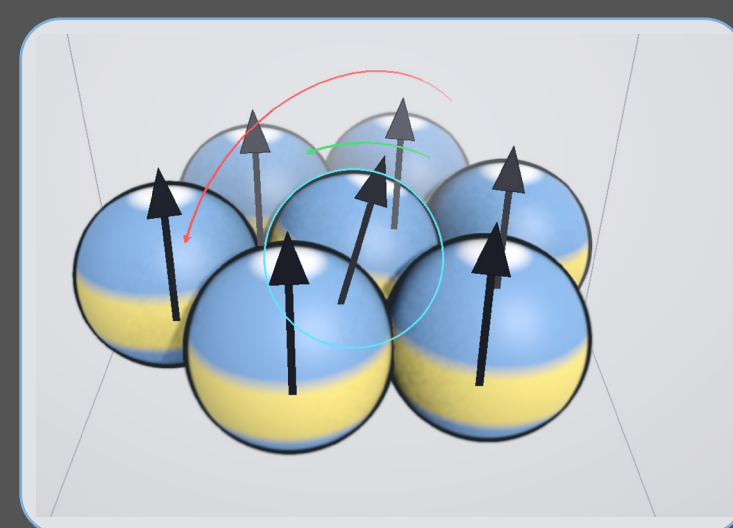
Interaction of two beads

2



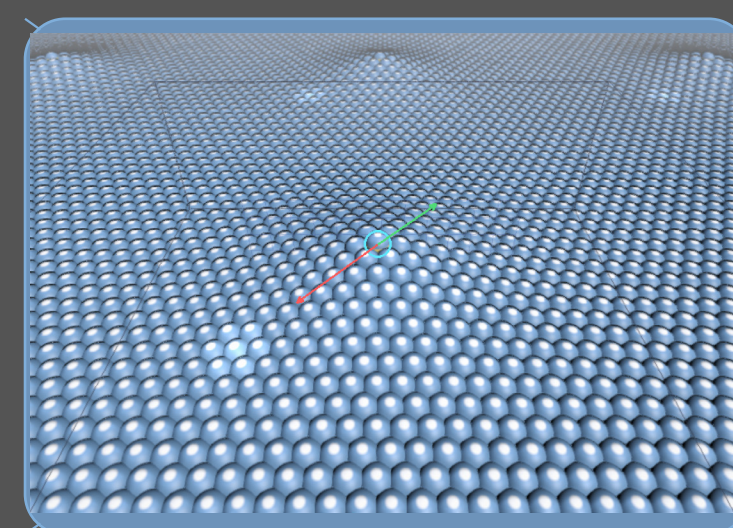
6+1 patch, feel the force

3



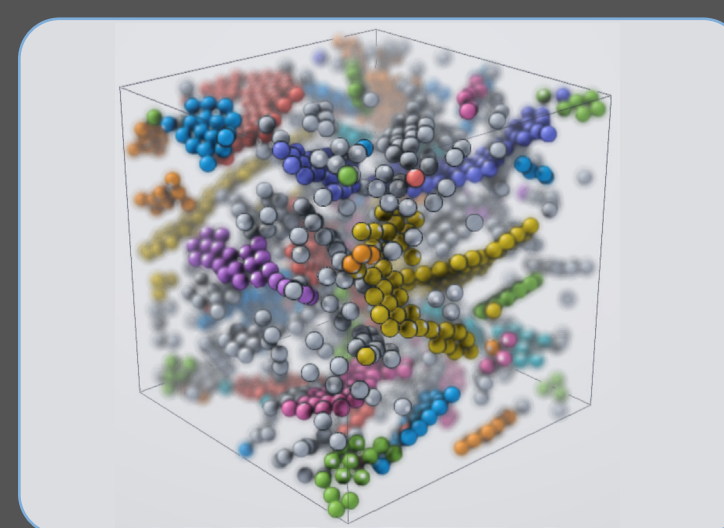
6+1 patch, feel the torque

4



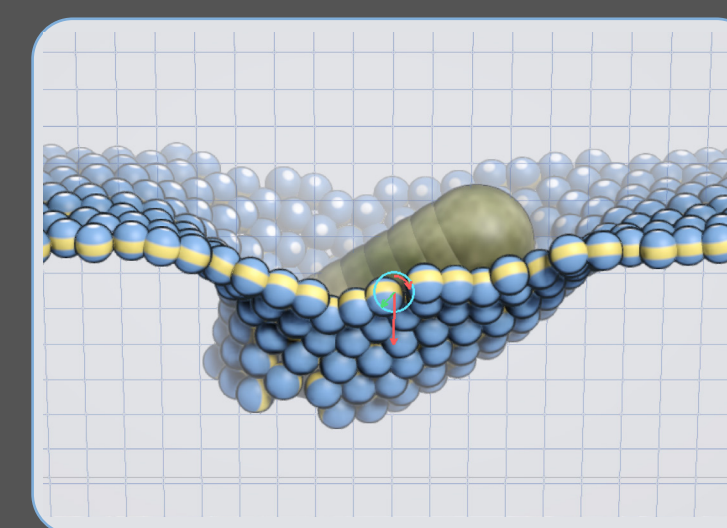
900 bead infinite membrane

5



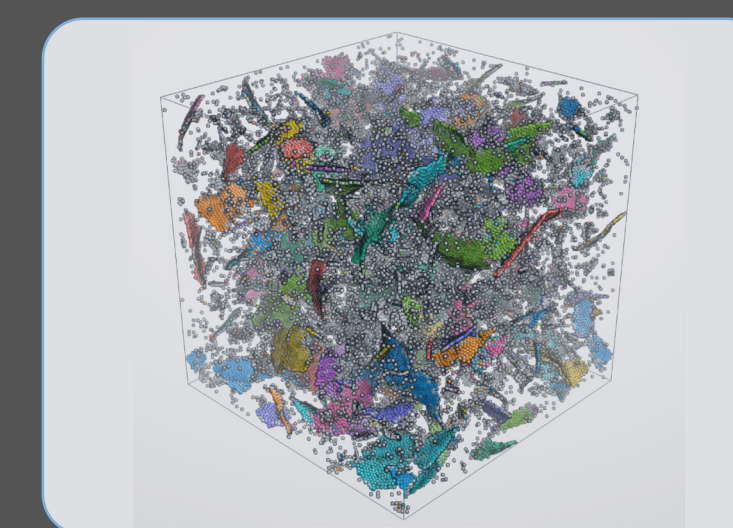
1500 beads in a box
--> self-assembly

6



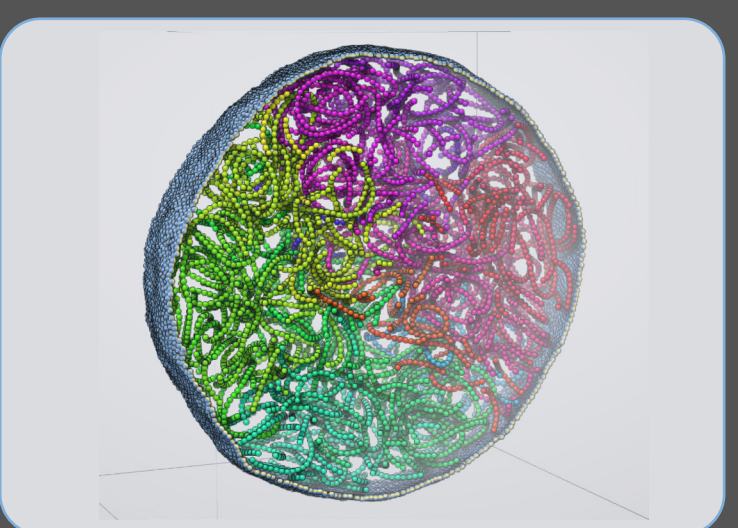
"endocytosis" of a rod
added rod-bead force term

7



50,000 beads on Snellius
--> large scale self-assembly

8



56,000 beads:
Membrane vesicle filled with
a polymer chain

Force feedback with a 25 year old joystick

- A 1998 Sidewinder joystick, run by our own Python driver (side-winder on GitHub).
- Python sets a spring force 60x a second; the stick's own chip pushes back far faster.
- The simulation's force moves the spring's centre and strength, to simulate a "pull" of your hand

```
from sidewinder.FF2 import FF2Device
ff = FF2Device() # open the stick over USB
s = ff.read_input() # read: s.x, s.y, s.twist, s.pressed
spring = ff.spring(stiffness=40) # start a force
# every frame: move the spring to the simulation's force
spring.set_condition(axis=0, cp_offset=fx,
                    pos_coef=k, neg_coef=-k)
ff.stop_all() # stop every force
```

Steering LAMMPS from Python

- LAMMPS runs inside Python, sharing memory with it.
- The joystick sets one extra force every frame.
- The simulation runs in short 20-step bursts to keep up with the screen.

```
lmp = LAMMPS() # LAMMPS as a library
lmp.command("variable fx internal 0.0") # a slot Python can write
lmp.command("fix drive probe addforce v_fx v_fy 0") # push the probe bead
lmp.set_internal_variable("fx", joystick_fx) # hand -> input force
lmp.command("run 20 pre no post no") # advance 20 steps
x = lmp.numpy.extract_atom("x") # positions, no copy
```

Live from the supercomputer (demo #7,8)

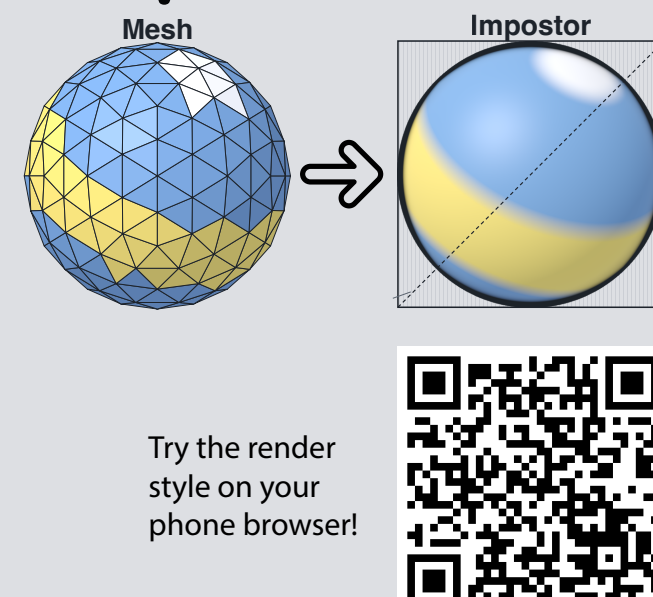
- LAMMPS runs on a Snellius GPU; the laptop only draws.
- TCP connection through an SSH tunnel, using Python's socket module.
- Each bead information is shrunk to the needed precision

make orientation vector 2D	quantize!	
	per bead	computed sent
	position	3 x 32 bit 3 x 12 bit
	orientation	3 x 32 bit 2 x 8 bit
	total	192 bit 52 bit

- 27 -> 7 MB/s for 56k beads at 20 fps, conference-wifi compatible...?

Real-time rendering of >50,000 spheres

- Each bead is a flat square; a shader computes a perfect sphere on it, pixel by pixel.
- All beads go to the GPU in one call, so 50,000 beads cost the CPU the same as ten.
- Shadows and ambient occlusion are added afterwards, at a cost per pixel, not per bead.



Try the render style on your phone browser!



Ask for more info :)

Get in contact

Consultancy/Collaborations

Visualisation & XR team:
Write visualisation@surf.nl

Get Compute on Snellius

<https://www.surf.nl/en/small-compute-applications-nwo>



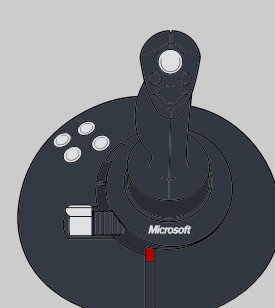
Learn new stuff!

- 2 week HPC school October 2026
- 1 day Blender course for Scientists on 28 Oct 2026
- Workshops, community events and more: <https://www.surf.nl/agenda>



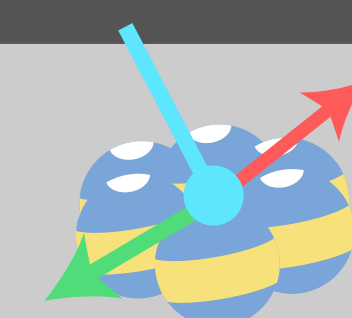
MesoMem

Sillano, Marrink & Idema, Phys. Rev. E (2026)



Joystick driver

github.com/stefanhuber1993/sidewinder



LAMMPS-live (mesomem-live)

github.com/stefanhuber1993/mesomem-live